PROJECT & TEAM · HEDERA CROSS CAMPUS CHALLENGE 2026

# EL JOKER

## 5ALIHA 3LINA. LEAVE IT TO US.



**The control layer for AI agents that spend money.** Five specialised agents buy for an organization: one AI proposes, four strict programs verify, decide and pay. Above the limit, the money waits in a Hedera vault that only a human co-signature can open. Every decision is fingerprinted on Hedera.

Multi-agent system · Claude (Anthropic) · Hedera Consensus Service · Scheduled tx · 2-of-2 vault · HTS · dinar token · Mirror Node verifier


TEAM
CHAIN619

MEMBERS
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PROBLEM

# THREE WAYS AI AGENTS FAIL WITH MONEY

01



## HALLUCINATE

Large language models produce fluent but unverified facts. An agent can invent a supplier, a price or an approval, then act on it with real money.

e.g. a “verified” broker with no license.

02



## OVERSPEND

Agents optimise the goal, not the budget. Retries, timeouts and race conditions turn one purchase into duplicate payments.

e.g. the same invoice paid twice after a network retry.

03



## HIDE THE TRAIL

Logs live in a database the operator controls. They can be edited or deleted after the fact: there is no independent audit trail.

e.g. a price changed after approval, and nobody can prove it.

**WHY IT MATTERS**

An NGO must justify every dinar to its donors. A company must respect its spending policy and pass its audits. An agent that spends money needs hard limits outside the model, and proof that nobody can rewrite.

## TODAY: AUTONOMY OR CONTROL. WE GIVE YOU BOTH.

SOLUTION · MEET THE CREW

# AI PROPOSES. RULES DECIDE. HEDERA PROVES.

Five specialised agents, one strict pipeline, a **separation of duties** like a real finance team. Only one agent uses AI, and it never touches the money.



## LEM3ALEM

**Coordinator · Plan**

Turns free text into a typed mission: who, where, what, budget. Schema-validated, no guessed facts.

## LEMHAF

**Research · the only AI**

Claude ranks the supplier offers and explains why. Strict JSON output. It can only recommend.

## NASNES

**Verifier**

10 independent checks: registry, license, evidence hashes, invoice reuse, prices, the AI's claims.

## 3ASES

**Guardian · no LLM**

13 fixed rules from the organization's policy, a risk score: approve, block or escalate.

## KANAWITA

**Payment · Hedera**

Pays only after an approval proven on Hedera. Idempotency key: never twice.

**CONTROL · A HUMAN** Above the limit, the money is locked in a 2-of-2 Hedera vault. Only the approver's co-signature releases it.

PRODUCT / POC · HOW A MISSION FLOWS

# REQUEST → PROOF

One mission, seven steps, five agents. Every handoff is visible in the chat, and every decision leaves a receipt on Hedera.

| REQUEST | PLAN | RESEARCH | VERIFY | GUARD | PAY | PROOF |
|---|---|---|---|---|---|---|
| **Operator** | **Lem3alem** | **Lemhaf · Claude** | **Nasnes** | **3ases** | **Kanawita** | **Hedera** |
| Types the need in plain language: who, where, budget, deadline. | Typed mission, validated. Missing facts are asked, never guessed. | Ranks the marketplace offers and explains the trade-offs. | 10 checks: registry, license, invoice reuse, prices, AI claims. | 13 fixed rules + risk score: approve, block or human review. | HTS transfer with an idempotency key. Paid exactly once. | Each decision fingerprinted (SHA-256) on HCS, chained. |

### AT EVERY STEP

The agent writes its decision to the database. EL JOKER hashes it (SHA-256 of canonical JSON), chains it to the previous hash and publishes it on the Hedera Consensus Service. Kanawita waits until the approval proof reaches consensus before paying.

### ABOVE THE LIMIT · HUMAN CO-SIGNATURE

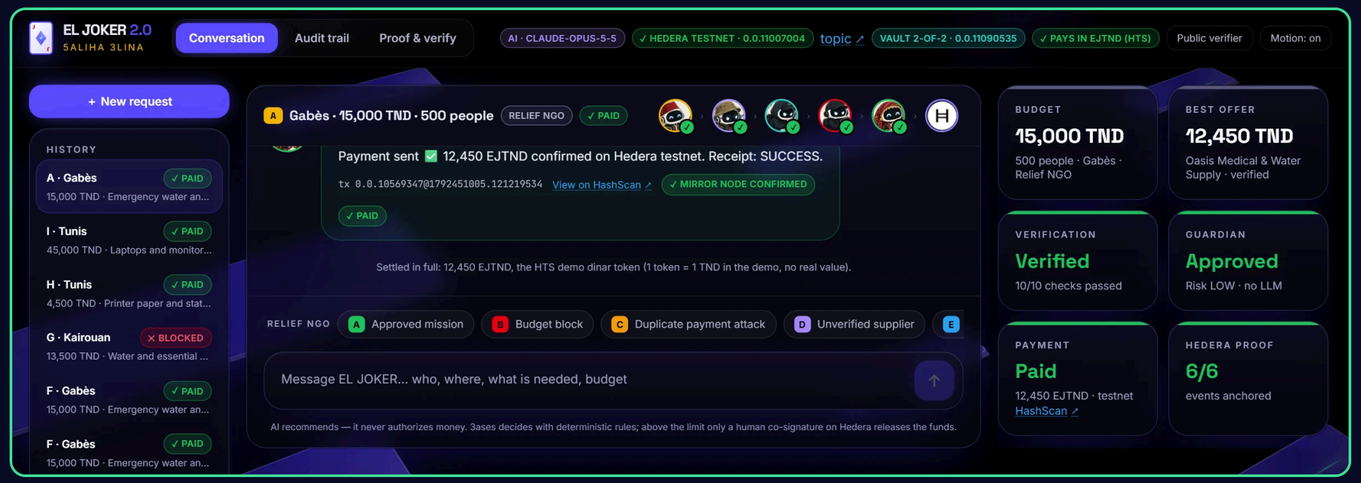
3ases escalates instead of approving. Kanawita creates a Hedera scheduled transfer from the vault, signed by the agents' key: 1 of 2. The money stays locked until the human approver signs: 2 of 2.

PRODUCT / POC · LIVE DEMO ON HEDERA TESTNET

# 15,000 TND BUDGET. MISSION CLEARED.

APPROVED

- **Lem3alem**: 500 people in Gabès, water and medicine, 15,000 TND, 48 h.
- **Lemhaf** (Claude) ranks 3 offers: **Oasis, 12,450 TND**, 20 h delivery.
- **Nasnes** 10/10 checks · **3ases** 12/12 rules, risk LOW, policy 2.1.0.
- **Kanawita**: HTS transfer of **12,450 EJTND**, receipt SUCCESS, confirmed by the Mirror Node.
- Fake supplier, budget too small, same payment twice: **blocked**, and the refusal is proven too.




Live run on Hedera testnet with Claude Opus 5.5: mission A, paid in EJTND, every proof anchored.

**6 / 6**
decisions anchored on HCS, chained by SHA-256

**12,450**
EJTND paid in full: 1 token = 1 dinar (test token)

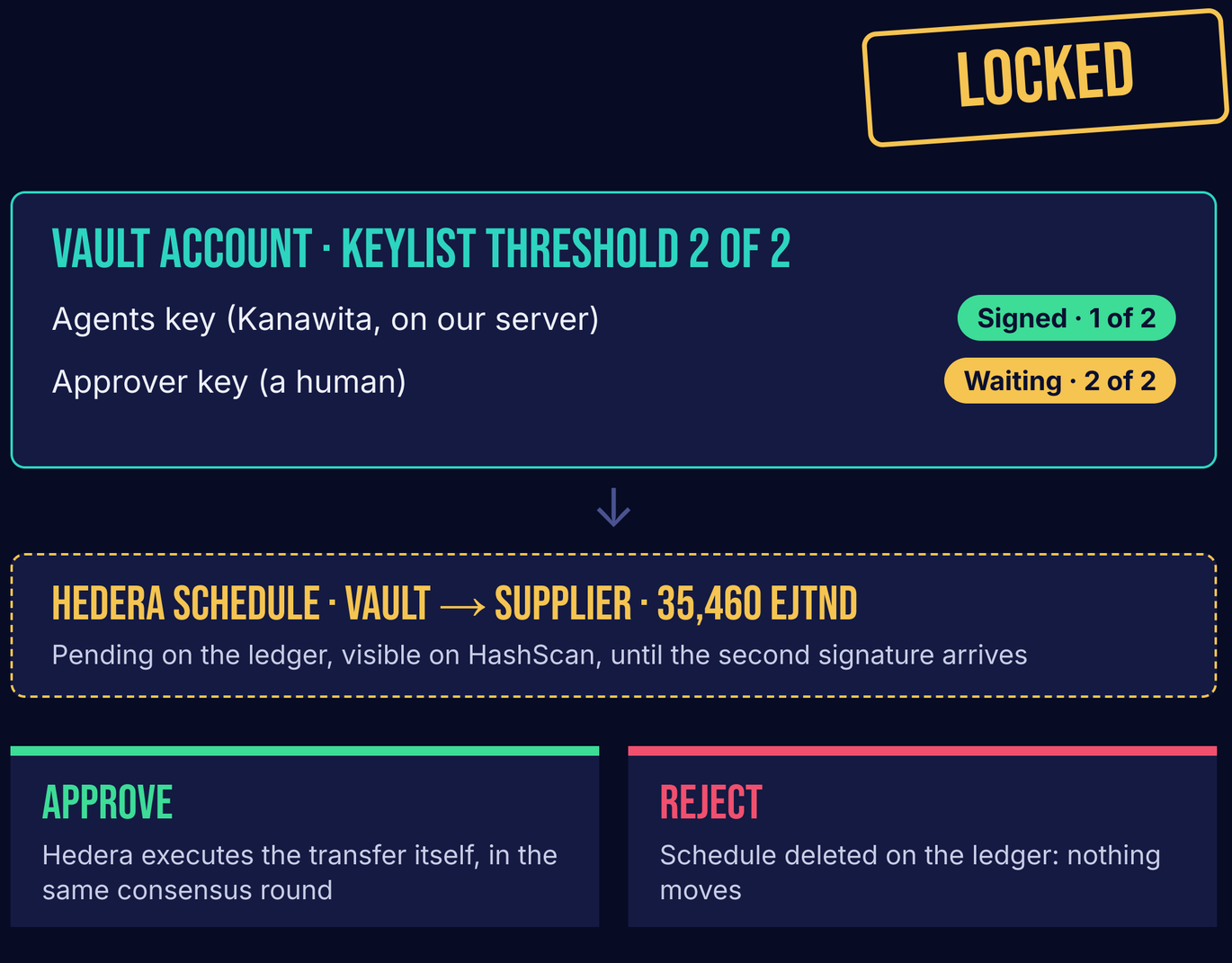
**1 PAYMENT**
idempotency key + database rule: never twice

PRODUCT / POC · HUMAN CO-SIGNATURE

# ABOVE THE LIMIT, HEDERA HOLDS THE MONEY.

- **ACME**, policy 1.1.0: autonomous limit 6,000 TND. Offer: **35,460 TND** of laptops.
- 3ases escalates. Kanawita can only **lock** it: **ScheduleCreateTransaction** from the vault, signed by the agents' key: 1 of 2.
- Operator + reason + PIN: the approver adds **ScheduleSignTransaction**: 2 of 2, and Hedera executes the transfer itself.
- Reject: **ScheduleDeleteTransaction**. The money never leaves the vault.

The approval is bound to the **input hash** of the exact facts reviewed: change one fact, and the approval is refused.




**Why it matters:** even if our server were hacked, its key is only one of two. The ledger, not our code, enforces the human signature.

TECHNOLOGY & HEDERA

# FOUR NATIVE HEDERA SERVICES



No smart contract to audit: native services **enforce the human signature**, carry the money and let anyone check. Finality in seconds, fees fixed in USD: a whole mission costs about one cent.

## CONSENSUS SERVICE

Every decision → SHA-256 of canonical JSON, chained to the previous hash, published with TopicMessageSubmitTransaction. Ordered, timestamped, immutable.

**Topic 0.0.11007004**

## SCHEDULED TX + KEYS

The vault's key is a KeyList with threshold 2 (agents + approver). ScheduleCreate → ScheduleSign → Hedera executes. ScheduleDelete on a rejection.

**Vault 0.0.11090535**

## TOKEN SERVICE

EJTND, our test dinar: 3 decimals like millimes, fixed supply, no admin key, so nobody can mint more. Paid in full with a TransferTransaction.

**Token 0.0.11090536**

## MIRROR NODE

Free REST history of topics, transactions and schedules. Our public verifier reads it from the browser and recomputes every hash with Web Crypto.

**testnet.mirrornode.hedera.com**

ARCHITECTURE · TYPESCRIPT MONOREPO

**WEB APP**
Next.js 16 + React, in the browser

→

**API · 5 AGENTS**
Express + SQLite; Claude for Lemhaf only

→

**HEDERA SDK**
HCS · schedules · HTS on testnet

→

**MIRROR NODE**
The full history, free to read

→

**PUBLIC VERIFIER**
/verify: any browser re-checks a mission

TECHNOLOGY & HEDERA · WHY NOT A DATABASE

# A DATABASE CAN REWRITE ITS PAST. HEDERA CAN'T.

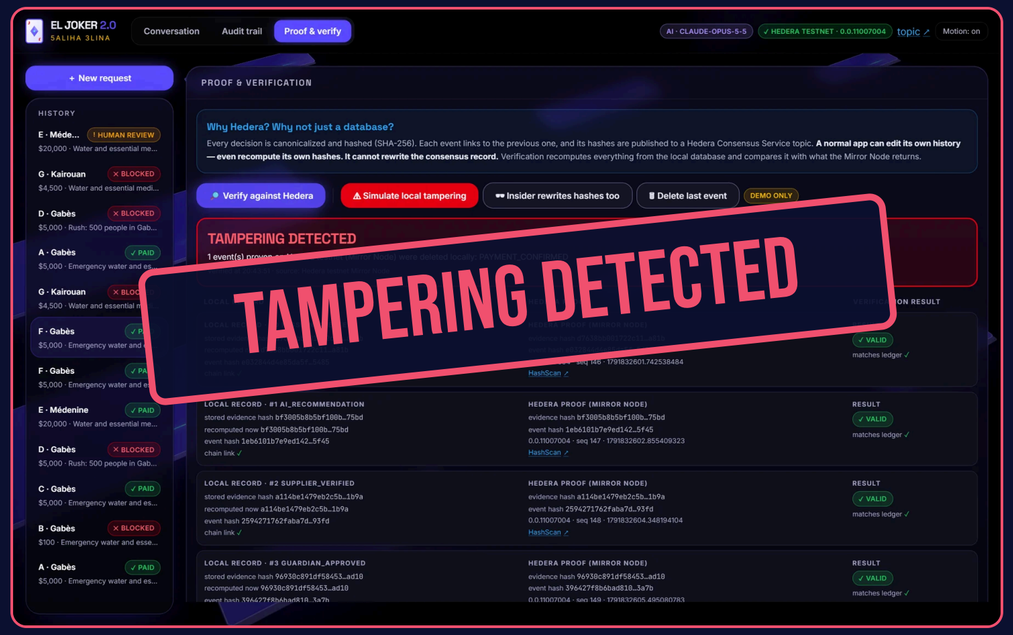
| Insider attack (tested live) | Local check | With Hedera |
|---|---|---|
| Edit the approved price | Caught | Caught |
| Edit it and recompute every hash | Missed | Caught |
| Delete the payment record | Missed | Caught |

## HOW THE PROOF WORKS

1. Each event is serialised as canonical JSON and hashed with **SHA-256**, together with the hash of the previous event: a chain.

2. The fingerprint is published on the **Hedera Consensus Service**: ordered, timestamped, final, impossible to edit.

3. **Verify** = recompute every hash from the database and compare it with the copy on the Mirror Node. One byte changed: **TAMPERING DETECTED**.




Local record vs Hedera proof vs verification result

Don't trust us: our **public verifier** runs in any browser, reads Hedera directly and needs no access to our servers.

IMPACT & ADOPTION

# WORKS TODAY. WHO NEEDS IT, AND WHY.



**9 / 9**

scenarios passed live on Hedera testnet, with real Claude

**115**

automated tests, including 7 in a real browser

**≈ 1 CENT**

of Hedera fees per mission, fixed in USD

**2**

organizations on one engine: an NGO and a company

FIRST

## NGOS & GRANTS

Emergency purchases that must be justified to donors.

**Value:** a public, tamper-proof audit trail per mission.

THEN

## COMPANIES

Agents in procurement, inside the spending policy.

**Value:** limits enforced by the ledger, faster purchases.

FOR EVERYONE

## DONORS & AUDITORS

Check any mission themselves, on Hedera.

**Value:** audits in minutes, without access to our servers.

PLUG IN

## AGENT PLATFORMS

Put EL JOKER in front of any AI agent that pays.

**Value:** rules, a human signature and proof as a service.

**Business model (our hypothesis):** a SaaS subscription per organization (policies, approvals, dashboard) plus a small fee per anchored proof. NGOs first, then companies.

**Honest limits today:** fictional suppliers, a test dinar token, an approver key held by our server, a demo PIN.

VISION

# EVERY AI AGENT THAT SPENDS MONEY, UNDER CONTROL.

Tomorrow, millions of AI agents will buy, book and pay. Each one should pass through **rules**, a **human for big amounts**, and **public proof**. EL JOKER wants to be that layer.

NOW

## PILOT-READY

Real suppliers and registries, a licensed partner for a real dinar token.

NEXT

## KEYS IN WALLETS

Approvers sign from their own wallet: our server holds only one key of two.

THEN

## TWO PILOTS IN TUNISIA

One NGO and one company, real purchases, measured results.

SCALE

## POLICY LIBRARY + SDK

Ready-made policies per sector, an SDK for any agent platform.

The trust layer for agentic payments on Hedera. Starting in Tunisia.



CONCLUSION

# AUTONOMOUS, BUT NEVER UNCONTROLLED.

### WHY IT MATTERS

AI agents now spend real money. Without hard limits and proof, no NGO or company can trust them.

### WHAT IS DIFFERENT

The AI never decides. Fixed rules decide, Hedera enforces the human signature above the limit, and anyone can verify every step.

### WHAT IS NEXT

Two pilots in Tunisia, approvers with their own wallets, a real dinar partner. Then any agent, any sector.

## EL JOKER — 5ALIHA 3LINA

THANK YOU

github.com/rihamghoudi-bit/el-joker-2 · hashscan.io/testnet/topic/0.0.11007004 · verify any mission: /verify